Yash Bavadiya

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EDUCATION

Indian Institute of Technology, Jodhpur **Aug 2024 – Jul 2028**
Bachelor's Degree - Computer Science and Engineering - **CGPA - 7.21** Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C++, C, Python, SQL
Core CS: Systems Design, Data Structures and Algorithms, Object-Oriented Programming, Software Design, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems
AI Engineering: Prompt Engineering, Retrieval-Augmented Generation, Agentic Pipelines, LLM Evaluation, Context Management, FAISS, Hugging Face, PyTorch
Technologies/Frameworks: Node.js, Express, React, Next.js, Qt (C++), Prisma, PostgreSQL, MySQL, MongoDB, Redis, WebSockets, Docker, Linux/UNIX, REST APIs, TCP/IP
Practices & Tools: Git, CI/CD (GitHub Actions), Pull Requests, Automated Testing, Test Coverage, Cross-Functional Collaboration, Agile/Scrum, Code Review, AI Coding Assistants, Nginx, Postman

EXPERIENCE

Google Summer of Code 2026 **May 2026 – Aug 2026**
Kdenlive @KDE Community | Open Source Contributor, C++ and Qt Remote
- Selected for **GSoC 2026** among 1,200 global contributors to design and ship Kdenlive's **RGB curve, gradient, and speed ramp editors** in **C++ and Qt**, owning features end to end from design through code review to release.
- Shipped **20 contributions** across Kdenlive and upstream repositories, incorporating technical feedback through rigorous code reviews to merge **high-quality pull requests** resolving an **11-year-old bug** now running for **41,000 monthly installs**.

Undergraduate Researcher **Jul 2025 – Nov 2025**
Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder IIT Jodhpur
- Architected an offline **Retrieval-Augmented Generation pipeline** in Python, integrating **FAISS vector search** with 4-bit quantized Phi-3 and Flan-T5-XL under strict context-window and memory constraints.
- Benchmarked **prompt engineering** and retrieval strategies across **8 ROUGE metrics** (**+71.77%** over baseline), reaching **3 second query latency** on 300-page PDFs for **15 users** across 3 feedback-driven iterations.

PROJECTS

Photon | GitHub | **C++20, Linux, UDP Multicast, Lock-Free Concurrency** **Jun 2026 – Jul 2026**
- Constructed **financial market infrastructure** and **systems design** from the wire up: a **NASDAQ ITCH 5.0 feed handler and matching engine** with zero-copy parsing, busy-polled UDP multicast, and a **lock-free SPSC ring**.
- Sustained **89k messages/sec** across **826 concurrent price-time priority order books** with zero hot-path heap allocation; verified by **27 GoogleTest cases**, **ASan/UBSan** presets, and clang-tidy CI.

Flock | GitHub | npm | **TypeScript, WebSocket, Redis, React, Docker** **Feb 2026 – Jun 2026**
- Released 3 npm packages enabling multiplayer cursors via **frontend technologies (React)** and a **10-line integration**, installable as a **55MB** self-hosted Docker image or npx.
- Implemented a horizontally scalable **Redis pub/sub relay** with keyspace-notification-driven eviction, securing **robust test coverage** with **58 automated tests** covering throttling, eviction, and delivery.

DaemonDoc (team of 2) | GitHub | Live | **TypeScript, LLM APIs, Express, MongoDB, Redis** **Jan 2026 – May 2026**
- Owned the **payments and billing system** end to end: designed **idempotent** webhook handling secured by **HMAC-SHA256** verification over **raw HTTP request bodies** before parsing.
- Built the **GitHub integration and prompt-construction layers** on a **Redis (BullMQ)** job queue with loop guards and optimistic concurrency, sustaining an **85% success rate** across **959 production runs** over 68 repositories for **50 users**.
- Cut per-run API cost roughly **10x** by routing file paths and token estimates, never file content, to a lightweight model for context selection, with failover across 6 provider keys.

COURSEWORK

- Data Structures and Algorithms
- Object-Oriented Programming
- Software Engineering
- Database Management Systems
- Operating Systems
- Machine Learning
- Pattern Recognition
- Probability and Statistics
- Discrete Mathematics

ACHIEVEMENTS

- Solved **450+** algorithmic problems on **LeetCode** in C++; Max Rating **1429** on Codeforces across rated contests.
- JEE Advanced 2024: AIR **3449** of 180,000+ candidates; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Merged **85+ open-source contributions** across GitHub and GitLab, shipping reviewed fixes to **Deno**, **Adobe React Spectrum**, **Kubernetes SIGs**, **nlohmann/json**, **Vitest**, and **Kdenlive**.

LEADERSHIP

- **Junior Core Team Member**, DevlUp Labs, IIT Jodhpur; prioritized hosting external open-source workshops for cross-functional stakeholders as the primary initiative, subsequently managing internal coding events and mentoring 30 students.
- **Core Member**, Programming Club and E-Cell, IIT Jodhpur; co-organized 5 competitive programming contests, 10 sessions, and 3 entrepreneurship events.